لبنان الغد

نموذج لمسابقة في مادة الرياضيات
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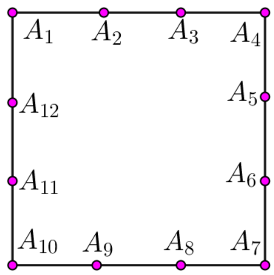
الصف: العلوم العامة	اللغة: الانكليزية	المدة: أربع ساعات

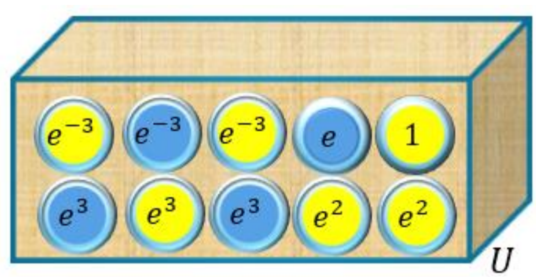
Question 1.

For each question, only one answer is true, find the correct one with **justification.**

Nº	Questions	Answer a	Answer b	Answer c
1.	The number of straight Lines passing through These 12 points is 	44	92	46
2.	Let $f(x) = x - \dfrac{2e^x}{1+e^x}$, $x \in IR$, then the slope, in an orthonormal system, of the tangent line at any point of (C_f) is always	greater than $\frac{1}{2}$	equal to $\frac{1}{2}$	strictly less than $\frac{1}{2}$
3.	Let $f(x) = x - 1 - \dfrac{2e^x+3}{1+e^x}$, $x \in IR$, then the reduced equation of the oblique asymptote to (C_f) at $+\infty$ is	$y = x - 1$	$y = -x + 1$	$y = x - 3$
4.	Let $f(x) = \dfrac{e^{2x} - 3e^x + 2}{1+e^x}$, $x \in IR$, then The curve (C_f), in an orthonormal system, is bellow the axis of abscissas when	$x > ln2$	$0 < x < ln2$	$x < 0$

Question 2:

An urn U contains ten balls indistinguishable by touch, holding The numbers **1**; $\boldsymbol{e}$; $\boldsymbol{e^2}$; $\boldsymbol{e^2}$; $\boldsymbol{e^3}$; $\boldsymbol{e^3}$; $\boldsymbol{e^3}$; $\boldsymbol{e^{-3}}$; $\boldsymbol{e^{-3}}$; $\boldsymbol{e^{-3}}$.



We select **successively without replacement** two balls from U.

Denote by **"a"** the number carried on the first drawn ball and by **"b"** the number carried on the second drawn ball.

Consider the following events:

A: "ln(ab)=0"; B: "ln(a)×ln(b)=0"; C: "$\dfrac{ln(a)}{\ln(b)}$=0"; D: "ln(b) < 0".

1) ***a***) Show that the probability of the event A is equal to $\frac{1}{5}$.

b) Prove that the two events A and B are **equiprobable**.

2) ***a***) Calculate the probability of the event C *and* show that the probability of D is equal to $\frac{3}{10}$.

b) Known that $\ln(b) < 0$. What is the probability that the event C will occur?

3) Calculate $P(C/B)$.

4) We designate by X the algebraic random variable equals to $lna + lnb$.

a) Verify that $P(X = 4) = 4/45$.

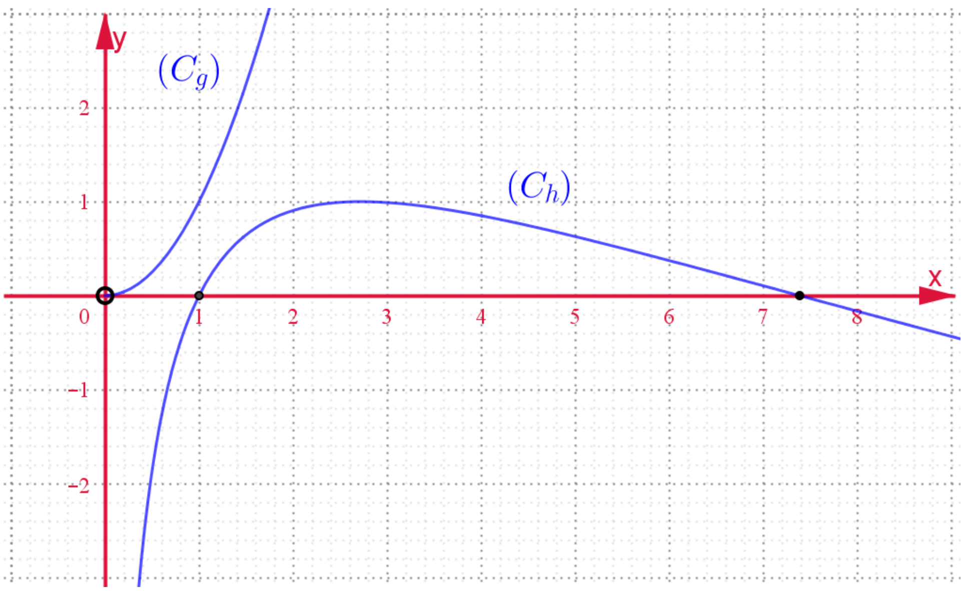
a) Determine the **eleven** values of X.

b) Calculate the probability of the event E: "$X \leq 4$".

Question 3:

Part A

The figure bellow represents the curves (C_g) and (C_h) of g and h , each defined over $]0;+\infty[$, respectively by: $g(x) = x^2$ and $h(x) = 2lnx - (lnx)^2$.



1) ***a***) Justify graphically that, for all $x \in]0;+\infty[, g(x) - h(x) > 0$.

b) Deduce that for all $x \in]0;+\infty[, \dfrac{2lnx - (lnx)^2}{x^2} < 1$.

2) ***a***) Verify that the function $K: x \to xlnx - x$ is a primitive on $]0;+\infty[$ of the function $x \to lnx$.

b) Deduce that $\int_1^{e^2} lnx.dx = 1 + e^2$.

c) Using integration by parts, show that $\int_1^{e^2} (lnx)^2.dx = 2e^2 - 2$.

d) Solve, in the interval $]0;+\infty[$, the equation $h(x) = 0$, and deduce the coordinates of the intersection points between (C_h) and $(x'x)$.

e) Deduce, in unit square, the area of the region bounded between (C_h) and $(x'x)$.

Part B

Consider the numerical function f defined over $]0;+\infty[$ by

$$f(x) = x - \frac{(lnx)^2}{x}$$

Designate by (C_f) the representative curve of the function f in an orthonormal system $(O;\vec{\imath};\vec{\jmath})$.

1) **a**) Verify that $\lim_{x\to 0^+} f(x) = -\infty$. Interpret graphically.

b) Verify that $\lim_{x\to+\infty} \frac{(lnx)^2}{x} = 0$. Calculate $\lim_{x\to+\infty} f(x)$

c) Deduce that the straight line $(d): y = x$ is an oblique asymptote of (C_f) in the $N(+\infty)$.

Prove that (d) is tangent to (C_f) at the point $A(1;1)$.

2) **a**) Show that for all $x \in]0;+\infty[, f'(x) = 1 - \frac{2lnx - (lnx)^2}{x^2}$.

b) Show that f is strictly increasing over $]0;+\infty[$. Set up its table of variation.

c) Show that there exists only one point B of (C_f) whose tangent line (T) is **strictly** parallel to (d).

d) Show that the equation $f(x) = 0$ has only one root β. Verify that $\beta \in]0.5;0.6[$ and that $\ln(\beta) = -\beta$

e) Plot in $(O;\vec{\imath};\vec{\jmath})$, the lines (d) and (T) and the curve (C_f).

3) ***a***) Show that the function f admits on $]0;+\infty[$ an inverse function f^{-1}

b) Determine the domain of definition of f^{-1}.

c) Prove that f^{-1} is differentiable at **zero** and that $(f^{-1})'(0) = \frac{\beta}{2+2\beta}$

d) Plot in the same system the curve $(C_{f^{-1}})$.

4) Let B' be the symmetric of B with respect to (d) and E the intersection of (d) with the straight line passing through B and perpendicular to $(x'x)$.

a) Show that the triangle BEB' is right isosceles at E.

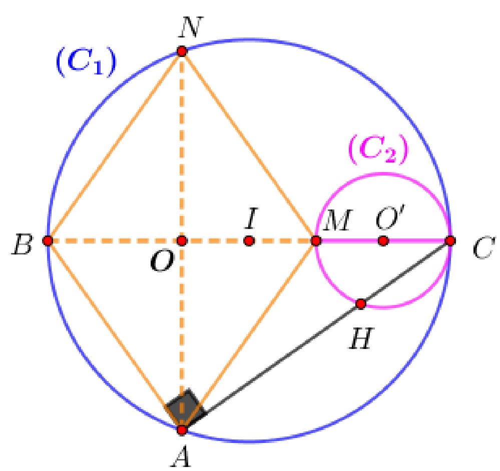
b) Verify that the area of the triangle BEB' is equal to $8 \times e^{-4}$ $(\text{unit})^2$.

b) Calculate, in unit square, the area of the region bounded between the two curves (C_f) and $(C_{f^{-1}})$ And the segment $[BB']$.

Question 4:

Based on the adjacent figure, consider the following information:

- $(\boldsymbol{C_1})$ is the circle of center I and dimeter $[BC]$, $BC = 6\ cm$
- M is a point of the segment $[BC]$ such that $BC = 3 \times MC$;
- O is the midpoint of the segment $[BM]$;
- $(\boldsymbol{C_2})$ is the circle of center O' and dimeter $[CM]$;
- A and N are two points on $(\boldsymbol{C_1})$ such that, $(\overrightarrow{AC};\overrightarrow{AB}) = \frac{\pi}{2}[2\pi]$ and $AMNB$ is a rhombus.
- (AC) cuts $(\boldsymbol{C_2})$ at the point H. $(H \neq C)$.



1) ***a***) Show that the two straight lines (HM) and (AB) are parallel.

b) Deduce that the three points H, M and N are collinear.

c) Show that $AB = 3 \times HM$ and $HA^2 = AB^2 - HM^2$.

2) Designate by S the direct plane similitude of center H that transforms A into M.

a) Precise the angle α of S and prove that its ratio $K = \frac{\sqrt{2}}{4}$.

b) determine the images of the two straight lines (AO) and (MH) under S. Deduce $S(N)$.

3) Show that $S(O) = O'$ and deduce that (HO) is tangent to $(\boldsymbol{C_2})$ at H.

4) Let $(\boldsymbol{C'_2})$ be image of $(\boldsymbol{C_2})$ under S Prove that $(O'H)$ is the tangent to the circle $(\boldsymbol{C'_2})$ at H.

5) The plane is referred to an orthonormal system $(O;\vec{u};\vec{v})$ with $\vec{u} = \overrightarrow{OI}$

a) Write the complex form of S.

b) Verify that the coordinates of the point H are $\left(8/3; -2\sqrt{2}/3\right)$.

c) Prove that the point D symmetric of M with respect to H belongs to (C_1).

Question 5: $\left((\boldsymbol{6\ points})\right)$

The complex plane (P) is referred to a **direct** orthonormal system $(O;\vec{u};\vec{v})$.

Let A, B and C be the points of respective affixes $i, 1+i$ and $-1+i$.

To each point M in the plane (P) of affix $z \neq i$, associates the point M' in the plane (P) of affix z'so that

$$z' = \frac{iz+2}{z-i}.$$

1) Show that for all $z \neq i$, $(z-i)(z'-i) = 1$.

2) Determine the set of points M in the plane (P) such that $M \equiv M'$.

3) **a)** Show that the points A, M and M' are collinear, if and only if, $(z-i)^2$ is real.

b) Deduce the set of points M in the plane (P) such that A, M and M' are collinear.

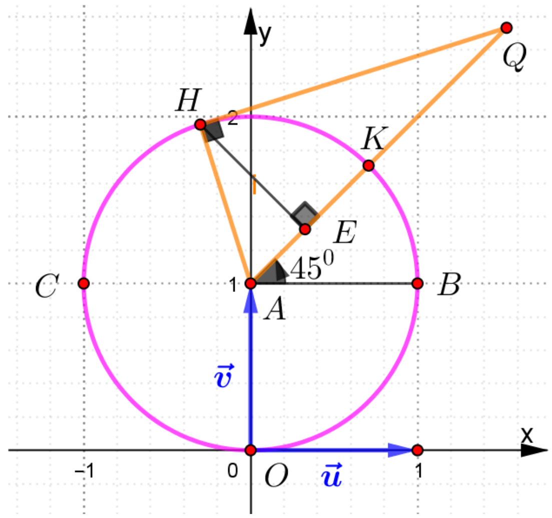
4) Show that for all M of affix $z \neq i$, $AM \times AM' = 1$ and $\left(\overrightarrow{AB};\overrightarrow{AM}\right) = \left(\overrightarrow{AM'};\overrightarrow{AB}\right)$ $[2\pi]$.

Based on the adjacent figure, consider the following information:

** (C) is the circle of center A and radius 1.

** K and Q are the points of affix $z_K = i + e^{i\frac{\pi}{4}}$ and $z_Q = i + re^{i\frac{\pi}{4}}, r > 1$.

** H is the point of (C) such that AHQ is a **right** triangle at $\boldsymbol{H}$ and E is the orthogonal projection of H on the straight line $(AK.)$



5) Let K' be the point of affix z'_K

a) Calculate the distance AK' and give a measure of $\left(\overrightarrow{AB};\overrightarrow{AK'}\right)$

b) Plot the point K' on the given figure.

c) Determine the set of points M' when M traces the semi straight line $[AK)$ except the point A

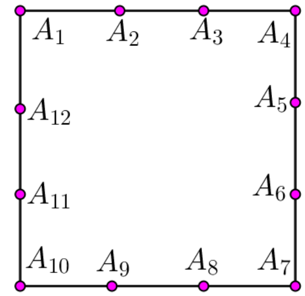
6) **a)** Show that $AQ \times AE = 1$.

b) Construct the point $Q'$of affix z'_Q

Wishing you all the best

Correction of the exam 2026. L.S.

Question 1:



1) The number of straight Lines passing through These 12 points is
$N = C_{12}^2 - 4 \times C_4^2 + 1 + 1 + 1 + 1 = 66 - 4 \times 6 + 4$
Then $N = 46$ and therefore, "$\boldsymbol{c}$" is true.

2) $f(x) = x - \dfrac{2e^x}{1+e^x}$, $x \in IR$.
f is differentiable over IR and for all real numbers,

$$f'(x) = 1 - 2\left(\frac{e^x(e^x+1) - e^x \times e^x}{(1+e^x)^2}\right) = \frac{1 + 2e^x + e^{2x} - 2e^x}{(1+e^x)^2} = \frac{1+e^{2x}}{(1+e^x)^2}$$

$$f'(x) - \frac{1}{2} = \frac{1+e^{2x}}{(1+e^x)^2} - \frac{1}{2} = \frac{2 + 2e^{2x} - 1 - 2e^x - e^{2x}}{2(1+e^x)^2} = \frac{1 - e^x + e^{2x}}{2(1+e^x)^2} = \frac{(1-e^x)^2}{2(1+e^x)^2} \geq 0$$

Therefore, the slope, in an orthonormal system, of the tangent line at any point of (C_f) is always greater than 0.5. therefore, "$\boldsymbol{a}$" is true.

3) $f(x) = x - 1 - \dfrac{2e^x + 3}{1+e^x}$, $x \in IR$

$$\lim_{x\to+\infty} [f(x) - (x-3)] = \lim_{x\to+\infty} \left(2 - \frac{2e^x+3}{1+e^x}\right) = 2 - 2 = 0.$$

Then, the reduced equation of the oblique asymptote to (C_f) at $+\infty$ is $y = x - 3$.
Therefore, "$\boldsymbol{c}$" is true.

4) Let $f(x) = \dfrac{e^{2x} - 3e^x + 2}{1+e^x}$, $x \in IR$, then,
The curve (C_f), in an orthonormal system, is bellow the axis of abscissas when
$e^{2x} - 3e^x + 2 < 0 \Rightarrow (e^x - 1)(e^x - 3) < 0 \Rightarrow 1 < e^x < 2 \Rightarrow 0 < x < ln2$. Therefore, "$\boldsymbol{b}$" is true.

Question 2:

Initial Data Analysis & Configuration

An urn U contains 10 balls which are identical and indistinguishable to the touch. We draw two balls sequentially and without replacement. Let $'a'$ be the number on the first ball drawn, and 'b' be the number on the second ball drawn. The distribution of the values and their respective natural logarithms is detailed below:

Ball Value (x)	Quantity in Urn	Natural Logarithm $\boldsymbol{ln}(x)$
1	1	$ln(1) = 0$
e	1	$ln(e) = 1$
e^2	2	$ln(e^2) = 2$
e^3	3	$ln(e^3) = 3$
e^{-3}	3	$ln(e^{-3}) = -3$

> *Total Number of Outcomes (Sample Space Ω):*
> *Since the drawing is successive and without replacement, the total number of ordered pairs is:*
> $Card(\Omega) = 10 \times 9 = 90$ *outcomes.*

1)

a) Show that the probability of event A is equal to $1/5$.

Event A is defined as: $'ln(ab) = 0'$. According to logarithmic identities, $ln(ab) = ln(a) + ln(b) = 0$, which yields $ln(b) = -ln(a)$. Let's determine the satisfying ordered pairs:

- **Case 1: One log value is 3 and the other is -3.**
  - Drawing e^3 first (3 choices), then e^{-3} (3 choices) → $3 \times 3 = 9$ outcomes.
  - Drawing e^{-3} first (3 choices), then e^3 (3 choices) → $3 \times 3 = 9$ outcomes.
- **Case 2: Both log values are $\mathbf{0}$ $(\ln(a) = 0 \; and \; \ln(b) = 0)$.**
  - This requires drawing the ball $'1'$ twice. Since the process is without replacement and there is only one ball $'1'$ in the urn, this case is impossible (0 outcomes).

$$\text{Total favorable outcomes} = 9 + 9 = 18$$
$$P(A) = 18/90 = 1/5$$

b) Prove that events A and B are equiprobable.

Event B is defined as: $'\ln(a) \times \ln(b) = 0'$. A product equals zero if and only if at least one factor is zero. This implies ln(a) = 0 or ln(b) = 0 (meaning at least one of the selected balls must be '1'):

- **Case 1: The first ball is $'1'$ ($\ln(a) = 0$) and the second is any remaining ball.**
  - Choices: $1 \times 9 = 9$ outcomes.
- **Case 2: The first ball is any other ball and the second ball is $'1'$ $(\ln(b) = 0)$.**
  - Choices: $9 \times 1 = 9$ outcomes.

$$\text{Total favorable outcomes} = 9 + 9 = 18$$
$$P(B) = 18/90 = 1/5$$

Since $P(A) = P(B) = 1/5$, both events are verified to be equiprobable.

2)

a) Calculate $P(C)$ and show that $P(D) = 3/10$.

- **Probability of Event C $('ln(a) \,/\, ln(b) = 0')$:**

A quotient is zero if and only if its numerator is zero and its denominator is non-zero. Hence, we require $\ln(a) = 0 \text{ AND } \ln(b) \neq 0$.
  - The first ball must be $'1'$ (1 choice).
  - The second ball must be any ball except $'1'$ (9 choices).
  - Favorable outcomes $= 1 \times 9 = 9$.

$$P(C) = 9/90 = 1/10.$$

- **Probability of Event D $('\ln(b) < 0')$:**

This implies that the second ball drawn must have a negative logarithm. The only eligible candidates are the 3 balls labeled e^{-3}. Let's count via sequential combinations:
  - Sub-case D1: First ball is an e^{-3} ball (3 choices), second ball is another e^{-3} ball (2 choices) → $3 \times 2 = 6$ outcomes.
  - Sub-case D2: First ball is NOT an e^{-3}ball (7 choices), second ball is an e^{-3} ball (3 choices) → $7 \times 3 = 21$ outcomes.
  - Total favorable outcomes $= 6 + 21 = 27$. $\quad$ $\mathbf{P(D) = 27/90 = 3/10}$.

b) Given that $ln(b) < 0$, what is the probability that event C is realized?

This requires evaluating the conditional probability $P(C|D) = P(C \cap D)/P(D)$.

The intersection event $C \cap D$ translates to: $\ln(a) = 0 \text{ AND } \ln(\mathrm{b}) < 0$.

- First ball must be $'1'$ (1 choice).
- Second ball must be one of the e^{-3} balls (3 choices).
- Favorable outcomes for $C \cap D = 1 \times 3 = 3$.

$P(C \cap D) = 3/90 = 1/30$.

$$\boldsymbol{P(C|D) = P(C \cap D)/P(D) = (3/90)/(27/90) = 3/27 = 1/9.}$$

c) Calculate $P(C|B)$.

We have, $P(C|B) = P(C \cap B)/P(B)$. Let's inspect the logical relationship between C and B:

Event C implies that $ln(a) = 0$ and $ln(b) \neq 0$. If C occurs, the product $ln(a) \times ln(b) = 0 \times ln(b) = 0$, which directly satisfies event B. Thus, event C is a subset of event B $(C \subset B)$, meaning $C \cap B = C$.

$$\boldsymbol{P(C|B) = P(C)/P(B) = (1/10)/(1/5) = 5/10 = 1/2.}$$

3) Calculer la probabilité que l'évènement $\boldsymbol{C}$ soit réalisé, sachant que l'un des évènements $\boldsymbol{B}$ ou $\boldsymbol{D}$ est réalisé.

$$* \boldsymbol{P\big(C\big|(B \cup D)\big) = \frac{P\big(C \cap (B \cup D)\big)}{P(B \cup D)} = \frac{P\big((C \cap B) \cup (C \cap D)\big)}{P(B \cup D)} = \frac{P(C \cap B) + P(C \cap D) - P(C \cap B \cap D)}{P(B) + P(D) - P(B \cap D)}}$$

* We Have calculated: $\boldsymbol{P(B) = 1/5 = 18/90, P(D) = 3/10 = 27/90, P(B \cap D) = 3/90}$

* $\boldsymbol{(C \cap D) \Rightarrow (lna = 0\ AND\ lnb < 0) \Rightarrow lna \times lnb = 0 \Rightarrow C \cap D \subset B \Rightarrow C \cap D \cap B = C \cap D}$

$$\boldsymbol{P(C \cap D) = P(C \cap B \cap D) \Rightarrow P(C \cap D) - P(C \cap B \cap D) = 0 \Rightarrow}$$

$$\boldsymbol{P\big(C\big|(B \cup D)\big) = \frac{P(C \cap B)}{P(B) + P(D) - P(B \cap D)} = \frac{\frac{9}{90}}{\frac{42}{90}} = \frac{9}{42} = \frac{3}{14}}$$

4) Let X be the algebraic random variable equal to $ln(a) + ln(b)$.

a) Verify that $P(X = 4) = 4/45$.

To attain a sum of $X = 4$, we evaluate the valid arrangements of ln values:

- **Case 1:** $\boldsymbol{ln(a) = 1 \text{ and } ln(b) = 3\ (ball\ 'e'\ then\ ball\ 'e^3')}$
  - Outcomes: 1 × 3 = 3.
- **Case 2:** $\boldsymbol{ln(a) = 3 \text{ and } ln(b) = 1\ (ball\ 'e^3'\ then\ ball\ 'e')}$
  - Outcomes: 3 × 1 = 3.
- **Case 3:** $\boldsymbol{ln(a) = 2\ and\ ln(b) = 2\ (\text{ball } 'e^2' \text{ then another ball } 'e^2')}$
  - Outcomes: 2 × 1 = 2 (since there are two e^2 balls, drawing without replacement is possible).

$$\textbf{Total outcomes for } \boldsymbol{(X = 4) = 3 + 3 + 2 = 8}$$

$$\boldsymbol{P(X = 4) = 8/90 = 4/45}$$

The targeted value is verified.

b) Determine the eleven values of X.

The possible scores for X are generated by compiling all permutations of two valid ln values from the available set $\{-3, 0, 1, 2, 3\}$ while adhering to structural availability constraints:

1. $(-3) + (-3) = -6 \quad (Valid\text{: } 3\ balls\ of\ e^{-3}\ exist)$
2. $(-3) + 0 = -3$
3. $(-3) + 1 = -2$
4. $(-3) + 2 = -1$
5. $(-3) + 3 = 0$

$6.0 + 1 = 1$

$7.0 + 2 = 2$

$8.0 + 3 = 3 \quad (or\ 1 + 2 = 3)$

$9.1 + 3 = 4 \quad (or\ 2 + 2 = 4)$

$10.2 + 3 = 5$

$11.3 + 3 = 6 \quad (Valid: 3\ balls\ of\ e^3\ exist)$

Hence, the set of the eleven possible values of X is:

$\boldsymbol{X(\Omega) = \{-6, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}}$

$\boldsymbol{c}$) Calculer la probabilité de l'évènement E : "$X \leq 4$".

$$P(X \leq 4) = 1 - P(X = 5) + P(X = 6) = 1 - \frac{2}{15} + \frac{1}{15} = \frac{12}{15} = \frac{4}{5}$$

Question 3 :

Part A

1) $\boldsymbol{a}$) Accorging to the figure, (C_g) is above (C_h) then, For all $x \in]0; +\infty[$, $g(x) - h(x) > 0$.

$\boldsymbol{b}$) Let $x \in]0; +\infty[, then\ g(x) - h(x) > 0$ this implies that $x^2 > 2lnx - (lnx)^2$.

Divide by the strictly positive real number x^2 we get'

$\frac{x^2}{x^2} > \frac{2lnx - (lnx)^2}{x^2}$, this leads to $\frac{2lnx - (lnx)^2}{x^2} < 1$.

2) $\boldsymbol{a}$) The function K is differentiable over $]0; +\infty[$. For $x \in]0; +\infty[, K(x) = xlnx - x$ then,

$K'(x) = 1 \times lnx + \frac{x}{x} - 1 = lnx + 1 - 1 = lnx$.

Therefore, K is a primitive on $]0; +\infty[$ of the function $x \to lnx$.

$\boldsymbol{b}$) $\int_1^{e^2} lnx.dx = (xlnx - x)\Big|_1^{e^2} = e^2lne^2 - e^2 - 0 + 1 = 2e^2 - e^2 + 1 = e^2 + 1$.

$\boldsymbol{c}$) $\int_1^{e^2} (lnx)^2.dx = \int_1^{e^2} (x)'(lnx)^2.dx = x(lnx)^2\Big|_1^{e^2} - \int_1^{e^2} \frac{2xlnx}{x}.dx = e^2(lne^2)^2 - 0 - \int_1^{e^2} 2lnx.dx$

$$\int_1^{e^2} (lnx)^2.dx = 4e^2 - 2(e^2 + 1) = 2e^2 - 2.$$

$\boldsymbol{d}$) Let $x \in]0; +\infty[$. $h(x) = 0 \Leftrightarrow 2lnx - (lnx)^2 = 0 \Leftrightarrow lnx(2 - lnx) = 0 \Leftrightarrow lnx = 0\ or\ lnx = 2$.

$$h(x) = 0 \Leftrightarrow x = 1\ or\ x = e^2.$$

Therefore, the two curves (C_h) and $(x'x)$ meet at the two points of coordinates $(1; 0)$ and $(e^2; 0)$.

$\boldsymbol{e}$) The area of the region bounded between (C_h) and $(x'x)$ is equal to

$\left(\int_1^{e^2} h(x).dx\right) \times (\text{unit})^2$ because for all $x \in [1; e^2], h(x) \geq 0$.

$$\int_1^{e^2} h(x).dx = \int_1^{e^2} [2lnx - (lnx)^2]dx = 2(e^2 + 1) - (2e^2 - 2) = 4.$$

Therefore, the required area is equal to $4 \times (\text{unit})^2$

Part B

$$x \in]0;+\infty[, f(x) = x - \frac{(lnx)^2}{x}$$

1) a) $\lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0^+}\left(x - \frac{(lnx)^2}{x}\right) = -\frac{\infty}{0^+} = (-\infty)\left(\frac{1}{0^+}\right) = (-\infty)(+\infty) = -\infty.$

Then, the straight line $(y'y): x = 0$ is a vertical asymptote to the curve (C_f).

b) $\lim\limits_{x\to+\infty}\frac{(lnx)^2}{x} = \frac{\infty}{\infty}$. Using H. R. we get, $\lim\limits_{x\to+\infty}\frac{(lnx)^2}{x} = \lim\limits_{x\to+\infty}\frac{\frac{2lnx}{x}}{1} = \lim\limits_{x\to+\infty}\frac{2lnx}{x} = 0.$

Calculate $\lim\limits_{x\to+\infty} f(x) = \lim\limits_{x\to+\infty}\left(x - \frac{(lnx)^2}{x}\right) = +\infty - 0 = +\infty.$

c) $\lim\limits_{x\to+\infty}(f(x) - x) = \lim\limits_{x\to+\infty}\left(x - \frac{(lnx)^2}{x} - x\right) = \lim\limits_{x\to+\infty}\frac{(lnx)^2}{x} = 0.$

Then, the straight line $(d): y = x$ is an oblique asymptote of (C_f) in the $N(+\infty)$.

Now, $M(x;y) \in (C_f) \cap (d) \Leftrightarrow M(x;y) \in (C_f)$ and $M(x;y) \in (d) \Leftrightarrow y = x - \frac{(lnx)^2}{x}$ and $y = x$

$x - \frac{(lnx)^2}{x} = x \Leftrightarrow \frac{(lnx)^2}{x} = 0 \Leftrightarrow (lnx)^2 = 0 \Leftrightarrow x = 1$ is a double root. For $x = 1, y = 1$

Therefore, the straight line (d) is tangent to (C_f) at the point $A(1;1)$.

2) a) The function f is differentiable over its domain of definition $]0;+\infty[$. And for all $x \in]0;+\infty[$,

$$f'(x) = 1 - \frac{x\left(\frac{2lnx}{x}\right) - 1 \times (lnx)^2}{x^2} = 1 - \frac{2lnx - (lnx)^2}{x^2}.$$

b) According to **Part A. 1) b):** we have, for all real number $x \in]0;+\infty[$,

$$\frac{2lnx - (lnx)^2}{x^2} < 1 \Rightarrow 1 - \frac{2lnx - (lnx)^2}{x^2} > 0 \Rightarrow f'(x) > 0.$$

Therefore, is strictly increasing over $]0;+\infty[$.

<u>table of variation of the function f over $]0;+\infty[$:</u>

x	0	$+\infty$
$f'(x)$	‖	$+$
$f(x)$	‖ $-\infty$	$\nearrow$ $+\infty$

c) Let $M(x;y) \in (C_f)$ such that the tangent line to (C_f) is strictly parallel to (d). This leads us to say that the slope of strictly parallel to (T) is equal to the slope of strictly parallel to (d).

$f'(x) = 1$ and $x \in]0;+\infty[- \{1\} \Leftrightarrow 1 - \frac{2lnx - (lnx)^2}{x^2} = 1 \Leftrightarrow \frac{2lnx - (lnx)^2}{x^2} = 0 \Leftrightarrow$

$2lnx - (lnx)^2 = 0 \Leftrightarrow lnx(2 - lnx) = 0 \Leftrightarrow lnx = 0$ or $lnx = 2 \Leftrightarrow x = 1$(rejected)or $x = e^2$.

$$x = e^2, y = e^2 - \frac{(lne^2)^2}{e^2} = e^2 - \frac{4}{e^2} = e^2 - 4e^{-2}.$$

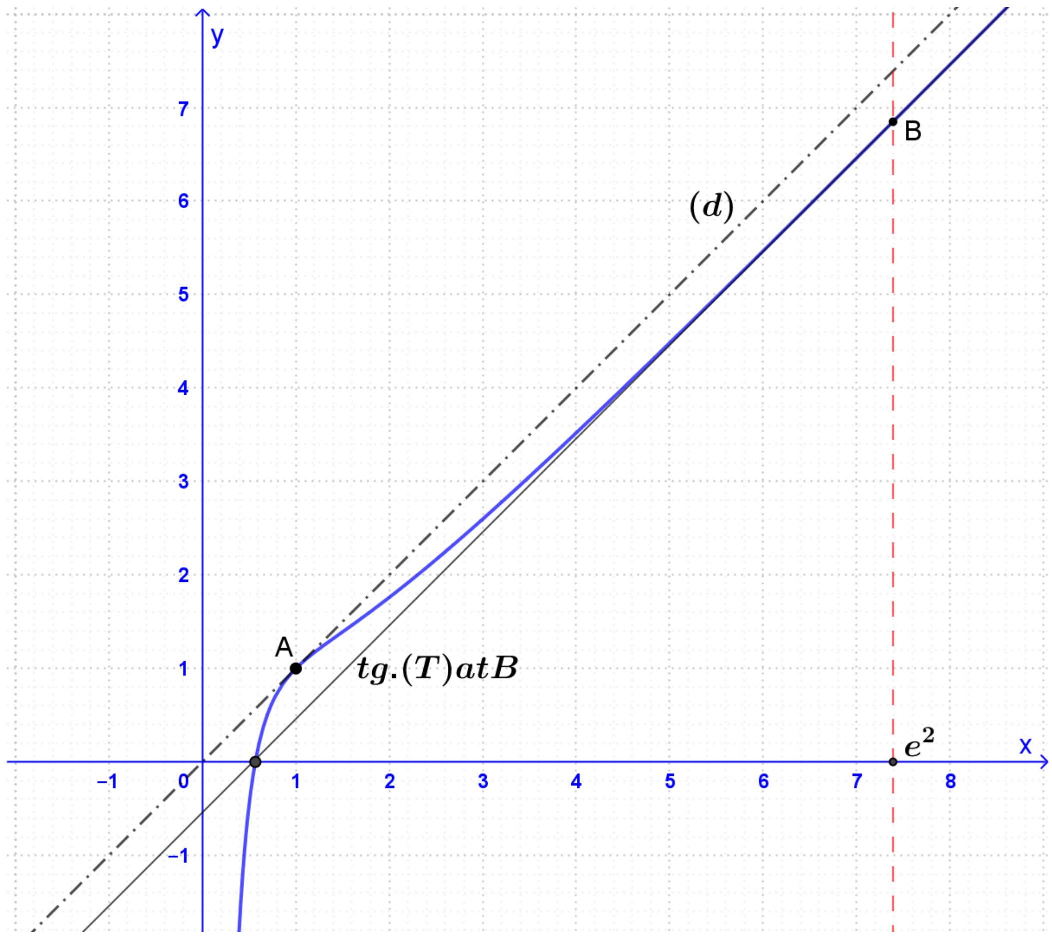
Therefore, there exists only one point $B(e^2; e^2 - 4e^{-2})$ of (C_f) whose tangent line (T) is strictly parallel to (d).

d) According to the table of variation of f: f is continuous, strictly increasing over $]0;+\infty[$ and it goes from negative to positive, this leads that the equation $f(x) = 0$ has only one root $\beta \in]0;+\infty[$.
0.5 and 0.6 $\in]0;+\infty[$ and $f(0.5) \times f(0.6) < 0$, then $\beta \in]0.5;0.6[$.

$$\mathrm{f}(\beta) = 0 \Leftrightarrow \beta - \frac{(ln\beta)^2}{\beta} = 0 \Leftrightarrow (ln\beta)^2 = \beta^2 \Leftrightarrow ln\beta = \beta \text{ or } ln\beta = -\beta.$$

But $ln\beta < 0$ because $\beta \in]0.5;0.6[$. Thus $\ln(\beta) = -\beta$.

e) Plot in $(O;\vec{\imath};\vec{\jmath})$ (d), (T) and the curve (C_f).



3) ***a***) The ft. f, on $]0;+\infty[$, is continuous and strictly increasing then, it has an inverse function f^{-1}.

b) The domain f^{-1} of is defined by $D_{f^{-1}} = f(]0;+\infty[) = \left]\lim_{x\to 0^+} f(x); \lim_{x\to +\infty} f(x)\right[$

Thus, $D_{f^{-1}} =]-\infty;+\infty[= IR$.

c) Prove that f^{-1} is differentiable at zero and that $(f^{-1})'(0) = \dfrac{\beta}{2+2\beta}$

Recall:

$\left(x \in D_{f^{-1}} = IR \text{ and } y = f^{-1}(x)\right) \Leftrightarrow \left(x \in D_f = \ ]0;+\infty[\text{ and } x = f(y)\right)$

$x = f(y)$. Differentiate with respect to x, we get:

$1 = y' \times f'(y)$, then $y' = \dfrac{1}{f'(y)}$, then $(f^{-1})'(x) = \dfrac{1}{f'(f^{-1}(x))}$ with $f^{-1}(x) \neq 0$

Property: if f is differentiable at a and $f'(a) \neq 0$ then, f^{-1} is differentiable at $b = f(a)$ and

$(f^{-1})'(f(a)) = \frac{1}{f'(\text{a})}$. $Or\ (f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$.

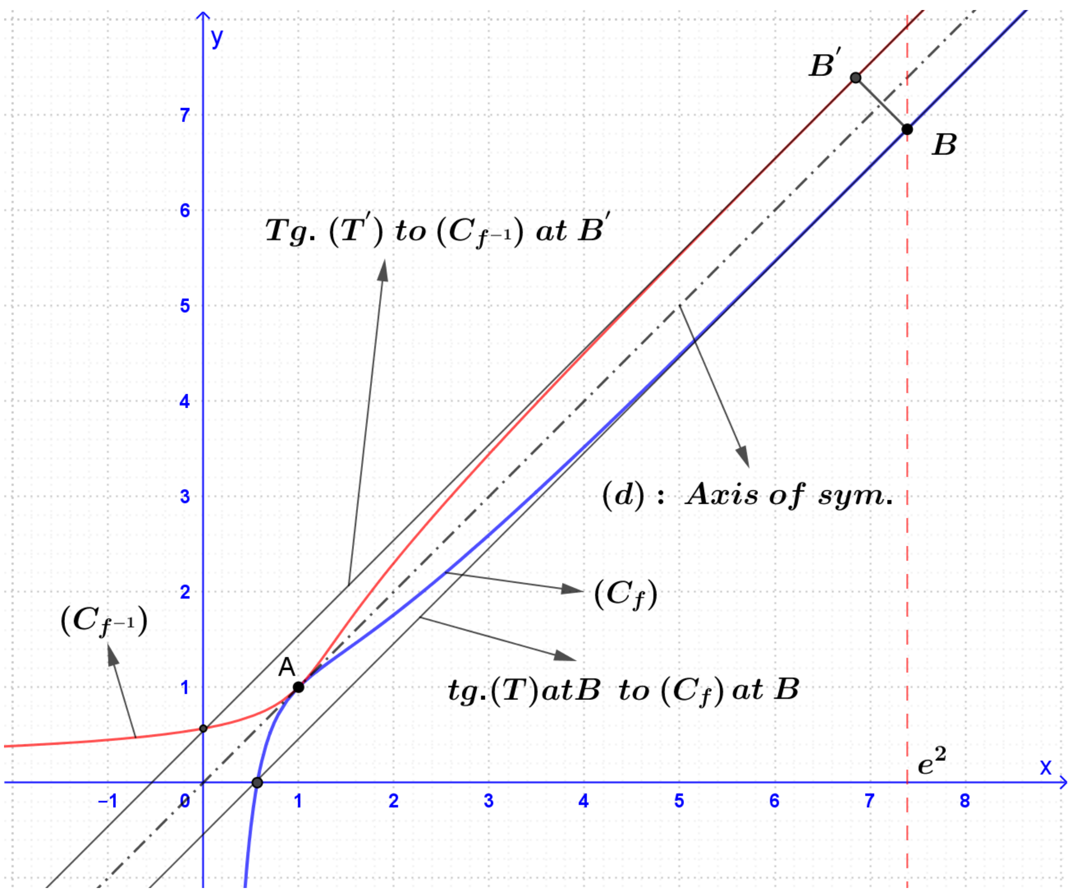
Now, f is differentiable at β and $f'(\beta) \neq 0$ then, f^{-1} is differentiable at $f(\beta) = 0$ and

$$(f^{-1})'(0) = \frac{1}{f'(f^{-1}(0))} = \frac{1}{f'(\beta)}.$$

$$f'(\beta) = 1 - \frac{2ln\beta - (ln\beta)^2}{\beta^2} = 1 - \frac{-2\beta - (-\beta)^2}{\beta^2} = \frac{\beta^2 + 2\beta + (-\beta)^2}{\beta^2} = \frac{2\beta^2 + 2\beta}{\beta^2} = \frac{2 + 2\beta}{\beta}$$

Then, $\frac{1}{f'(\beta)} = \frac{\beta}{2 + 2\beta}$. Therefore, $(f^{-1})'(0) = \frac{\beta}{2 + 2\beta}$

d) Plot in the same system the curve $(C_{f^{-1}})$.



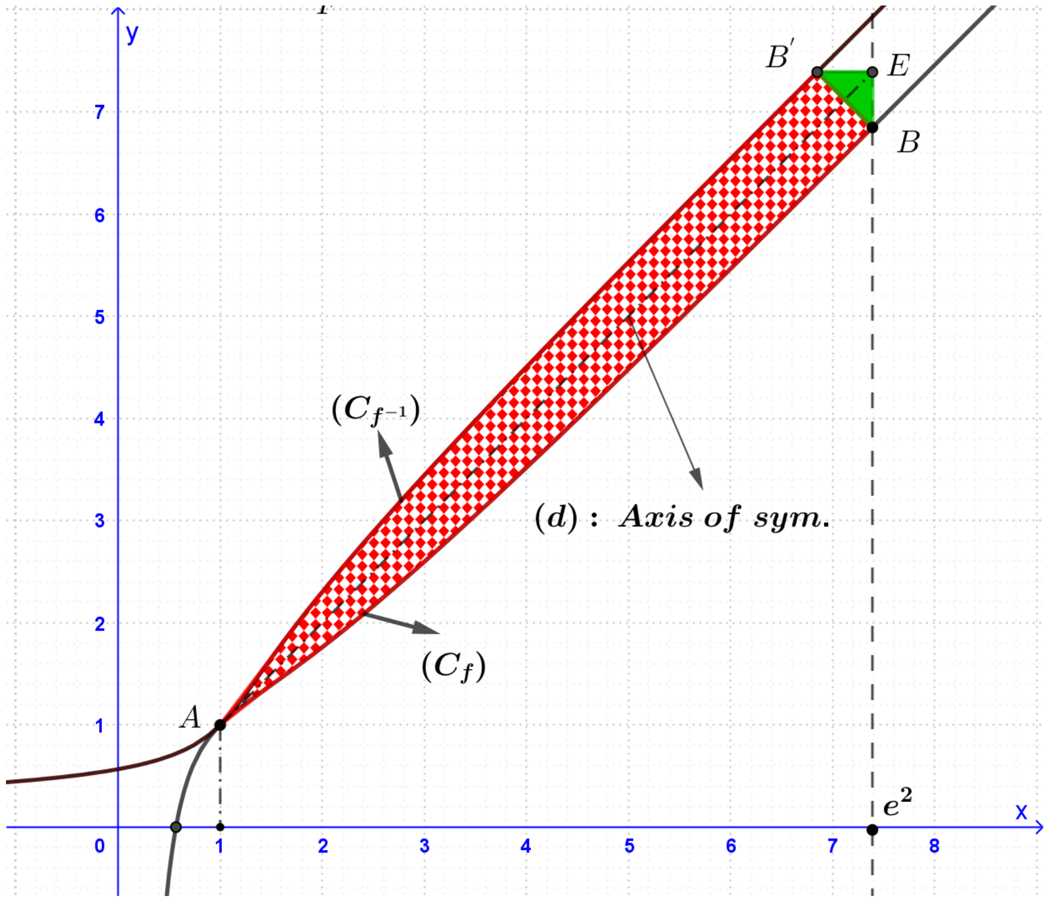
- The two curves (C_f) and $(C_{f^{-1}})$ are symmetric with respect to the straight line $(d): y = x$.
- (d) is the oblique asymptote to (C_f) in the $N(+\infty)$ then, (d) is the oblique asymptote to $(C_{f^{-1}})$ in the $N(+\infty)$
- (d) is tangent to (C_f) at $A(1; 1)$, then (d) is tangent to $(C_{f^{-1}})$ at $A' \equiv A(1; 1)$.
- (T) is parallel to (d) and tangent to (C_f) at $B(e^2; e^2 - 4e^{-2})$, then (T') is parallel to (d) and tangent to $(C_{f^{-1}})$ at $B'(e^2 - 4e^{-2}; e^2)$.

4) ***a***) We have, $E(e^2; e^2) \in (d): y = x; B(e^2; e^2 - 4e^{-2}) \in (C_f)\ ; B'(e^2 - 4e^{-2}; e^2)$

$$\overrightarrow{EB}(0; -4e^{-2});\ \overrightarrow{EB'}(-4e^{-2}; 0).$$

$$\overrightarrow{EB}.\overrightarrow{EB'} = 0 = 0 = 0\ and\ \|\overrightarrow{EB}\| = 4e^{-2} = \left\|\overrightarrow{EB'}\right\|$$

Therefore, the triangle BEB' is right isosceles at E.



$\boldsymbol{b}$) Area of the triangle BEB' is equal to $4e^{-2} \times 4e^{-2} \times 0.5\ (\text{unit})^2 = 8 \times e^{-4} \times\ (\text{unit})^2$
Therefore, the area of the triangle BEB' is equal to $8 \times e^{-4} \times\ (\text{unit})^2$.
$\boldsymbol{b}$) Let $\boldsymbol{S}$ be the area of the region bounded between the two curves (C_f) and $(C_{f^{-1}})$ and the segment $[BB']$. Then,

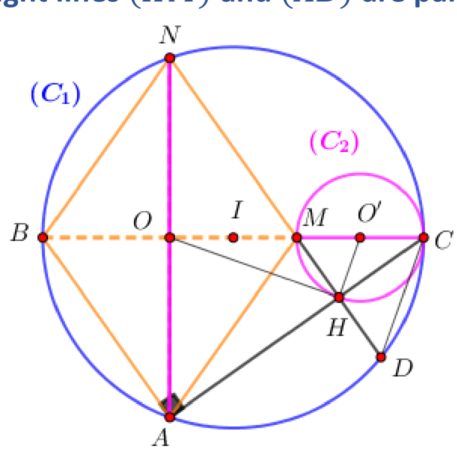
$$\boldsymbol{S} = \left(2\int_1^{e^2} \left(y_{(d)} - f(x)\right) dx - 8 \times e^{-4}\right) \times\ (\text{unit})^2 = \left(\frac{16}{3} - \frac{8}{e^4}\right) \times\ (\text{unit})^2, \text{because,}$$

$$2\int_1^{e^2} \left(y_{(d)} - f(x)\right) dx = 2\int_1^{e^2} \left(y_{(d)} - f(x)\right) dx = 2\int_1^{e^2} \frac{(lnx)^2}{x} dx = 2\int_1^{e^2} (lnx)'(lnx)^2 dx = \frac{16}{3}.$$

Question 4:

1) Geometric Properties

a) Show that the two straight lines (HM) and (AB) are parallel.



- Angle$\angle \boldsymbol{BAC}$: The point A lies on the circle (C_1) with diameter $[BC]$.
  An angle inscribed in a semicircle is a right angle, so $\angle BAC = 90°$. This implies that $(AB) \perp (AC)$.
- Angle $\angle \boldsymbol{MHC}$: The point H lies on the circle (C_2) with diameter $[CM]$. Similarly, the inscribed angle facing its diameter is a right angle, so $\angle MHC \ = \ 90°$. This implies that $(HM) \perp (AC)$.
- **Conclusion:** Since both straight lines (AB) and (HM) are perpendicular to the same straight line (AC), they must be parallel to each other: $(\boldsymbol{HM}) \parallel (\boldsymbol{AB})$.

b) Deduce that the three points $\boldsymbol{H}, \boldsymbol{M}$, and $\boldsymbol{N}$ are collinear.

- Since $AMNB$ is a rhombus, its opposite sides are parallel: $(\boldsymbol{NM}) \parallel (\boldsymbol{AB})$
- From part **1) a**, we have already proven that: $(\boldsymbol{HM}) \parallel (\boldsymbol{AB})$
- According to Euclid's parallel postulate, through a given point M not on the line (AB), there exists only one unique line parallel to (AB).
- Since both lines (NM) and (HM) pass through M and are parallel to (AB), they must be the exact same straight line. Thus, the points $\boldsymbol{H}, \boldsymbol{M}$, **and $\boldsymbol{N}$ are collinear.**

c) Show that $\boldsymbol{AB} = 3 \times \boldsymbol{HM}$ and $\boldsymbol{HA}^2 = \boldsymbol{AB}^2 - \boldsymbol{HM}^2$.

- **Proving $\boldsymbol{AB} = \mathbf{3} \times \boldsymbol{HM}$:**
  In triangle CAB, the line (HM) is parallel to (AB). Applying Thales's Theorem (or triangle similarity $\Delta CHM \sim \Delta CAB$), (congruent triangles) we get the side ratios:
  $$HM/AB = CM/CB$$
  We are given that $BC = 3 \times MC$, which means $CM/CB = 1/3$. Substituting this into the ratio yields:
  $$\boldsymbol{HM/AB} = \mathbf{1/3} \Rightarrow \boldsymbol{AB} = \mathbf{3} \times \boldsymbol{HM}$$
- **Proving $\boldsymbol{HA}^2 = \boldsymbol{AB}^2 - \boldsymbol{HM}^2$:**
  Since H lies on the line segment $[AC]$ and $(MH) \perp (AC)$, the triangle AHM is a right-angled triangle at H.
  Applying the Pythagorean Theorem in ΔAHM:
  $$HA^2 + HM \ = AM^2$$
  Because $AMNB$ is a rhombus, all its sides are equal in length, meaning $AM = AB$. Substituting AB for AM gives:
  $$\boldsymbol{HA^2} + \boldsymbol{HM^2} = \boldsymbol{AB^2} \Rightarrow \boldsymbol{HA^2} = \boldsymbol{AB^2} - \boldsymbol{HM^2}.$$

2) Direct Plane Similitude S

a) Precise the angle α of S and prove that its ratio $\boldsymbol{K} = \sqrt{2}/4$.

- **Angle $\boldsymbol{\alpha}$:**
  The direct similitude S has center H and transforms A into M, $(S(A) \ = \ M)$. The angle α is defined by the directed vector angle: $\alpha = (\overrightarrow{HA}; \overrightarrow{HM})[\text{mod } 2\pi]$
  Since ΔAHM is right-angled at H, the lines are perpendicular. Following the clockwise (negative) orientation from $\overrightarrow{HA}$ to $\overrightarrow{HM}$ based on the given geometric setup $(\overrightarrow{AC}; \ \overrightarrow{AB}) \ = \ \pi/2$, we find:
  $$\boldsymbol{\alpha} = -\boldsymbol{\pi}/\mathbf{2} \ [\boldsymbol{mod} \ \mathbf{2}\boldsymbol{\pi}]$$
- **Ratio $\boldsymbol{K}$:**
  The ratio is defined as $K = HM/HA$. From part 1) c, we know that $AB = 3HM$, so $AB^2 = 9HM^2$.
  Substituting this into the Pythagorean identity:
  $$HA^2 = AB^2 - HM^2 = 9HM^2 - HM^2 = 8HM^2$$
  Taking the square root on both sides:

$$HA = \sqrt{8}\,HM = 2\sqrt{2}\,HM$$

Now, calculating the scaling ratio K:

$$\boldsymbol{K = HM/(2\sqrt{2}HM) = 1/2\sqrt{2} = \sqrt{2}/4}$$

b) Determine the images of the two straight lines (AO) and (MH) under S. Deduce $S(N)$.

- **Image of (AO):** under S ?

We have $S: A \to M$, then the image of (AO) under S is a straight line passing through M and perpendicular to (AO). Thus, $\boldsymbol{S(AO) = (MO)}$.

- **Image of (MH) under S ?**

We have $S: H \to H$, then the image of (MH) under S is a straight line passing through H and perpendicular to (MH). Thus, $\boldsymbol{S\big((MH)\big) = (HC)}$.

- We have $S: \begin{cases} AO \to MO \\ MH \to HC \end{cases}$ and $(AO) \cap MH = \{N\}$ and $(MO) \cap HC = \{C\}$, then $\boldsymbol{S(N) = C}$.

3) Midpoints and Tangency

Show that $\boldsymbol{S(O) = O'}$: O is the midpoint of $[AN]$ and $S: \begin{cases} A \to M \\ N \to C \end{cases}$ then, $S(O) = O'$ which is the midpoint of $[MN]$. the midpoint O directly maps to the corresponding midpoint $\boldsymbol{O'}$.

Deduce Tangency: $S: \begin{cases} O \to O' \\ H \to H \end{cases}$ and $\alpha = -\pi/2$, then (HO) is perpendicular to (HO')

H belongs to (C_2) and HO' is a radius of (C_2), **Therefore, the line $\boldsymbol{(HO)}$ is tangent to $\boldsymbol{(C_2)}$ at $\boldsymbol{H}$.**

4) Image of a Circle and Tangency

- Similitudes perfectly preserve geometric properties such as tangency and angles.
- From Part 3, we established that the straight line (HO) is tangent to the circle (C_2) at point H.
- Let's look at the images of these elements under S:
  - The image of the circle (C_2) *is* (C'_2).
  - The image of the point of tangency H is $S(H) = H$.
  - The image of the line (HO) is the line (HO') (since $S(H) = H$ and $S(O) = O'$).
- Since the transformation preserves the contact property, the image line (HO') must be tangent to the image circle (C'_2) at the image point H.
- Thus, the line $\boldsymbol{(O'H)}$ **is tangent to the circle $\boldsymbol{(C'_2)}$ at $\boldsymbol{H}$.**

5) Complex form of a similitude

a) S is a direct plane similitude of ration $\sqrt{2}/4$ and angle $-\pi/2$,its complex form is given as :

$z' = \dfrac{\sqrt{2}}{4}e^{-i\frac{\pi}{2}}z + b$, $O(0,0) \to O'(3,0)$. $z_{O'} = \dfrac{\sqrt{2}}{4}e^{-i\frac{\pi}{2}}z_O + b$; leads to $b = 3$

therefore , $z' = -\dfrac{\sqrt{2}}{4}iz + 3$.

$\boldsymbol{b})\ z_H = \dfrac{b}{1-a} = \dfrac{3}{1+\dfrac{\sqrt{2}}{4}} i = \dfrac{8}{3} - \dfrac{2\sqrt{2}}{3} i \to H\left(\dfrac{8}{3}\,;-\dfrac{2\sqrt{2}}{3}\right)$

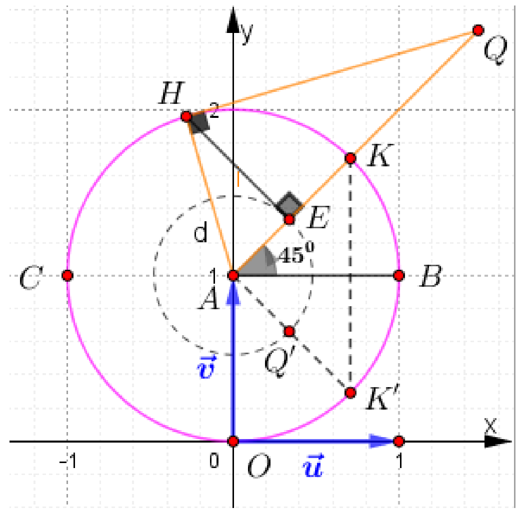
$\boldsymbol{c})\ z_H = \dfrac{z_M + z_D}{2} \to z_D = 2z_H - z_M = \dfrac{16}{3} - \dfrac{4\sqrt{2}}{3} i - 2 = \dfrac{10}{3} - \dfrac{4\sqrt{2}}{3} i$

$ID = |z_D - z_I| = \left|\frac{10}{3} - \frac{4\sqrt{2}}{3} i - 1\right| = \left|\frac{7}{3} - \frac{4\sqrt{2}}{3} i\right| = \sqrt{\left(\frac{7}{3}\right)^2 + \left(\frac{4\sqrt{2}}{3}\right)^2} = \sqrt{9} = 3 = \ radius\ of\ (C_1).$

Therefore $\boldsymbol{D} \in (\boldsymbol{C_1})$

Question 5:

Verification of the Fundamental Algebraic Relation



We want to show that for all $z \neq i, (z-i)(z'-i) = 1.$

Substituting the explicit expression for z' into the left-hand side:

$(z - i)(z' - i) = (z - i) \times [\,(iz+2)/(z-i) - i\,]$

By finding a common denominator within the brackets, we get:

$= (z - i) \times [\,(iz + 2 - i(z - i))\, /\, (z-i)\,]$

Since $z \neq i$, the term $(z-i)$ non-zero and cancels out cleanly:

$(z - i)(z' - i) = iz + 2 - i(z - i)$

$= iz + 2 - iz + i^2$

$= 2 + (-1) = 1$

Hence, the relation holds true: $(\boldsymbol{z} - \boldsymbol{i})(\boldsymbol{z}' - \boldsymbol{i}) = \mathbf{1}.$

2) Deduction of the Non-Inclusivity Condition ($\boldsymbol{z}' \neq \boldsymbol{i}$)

From the identity proven in part (1), we have $(z-i)(z'-i) = 1.$

Let us perform a proof by contradiction. Suppose there exists a valid point $z \neq i$ such that its image satisfies $z' = i$. This would immediately imply that:

$$(z' - i) = 0$$

Substituting this into our primary relation yields:

$$(z-i) \times 0 = 1 \Rightarrow 0 = 1.$$

This is an absolute mathematical contradiction. Therefore, our original assumption must be false.

Consequently, for all $\boldsymbol{z} \neq \boldsymbol{i}$, it is structurally impossible for the image to equal $\boldsymbol{i}$, meaning $\boldsymbol{z}' \neq \boldsymbol{i}$.

3) Identification of Invariant Points ($\boldsymbol{M} \equiv \boldsymbol{M}'$)

To find the invariant or fixed points of this transformation, we solve the equation $M \equiv M'$, which translates to setting their complex affixes equal: $z = z'$.

$$z = (iz+2)/(z-i)$$

Cross-multiplying by $(z-i)$ since $z \neq i$ gives:

$$z(z-i) = iz + 2 \Leftrightarrow$$

$$z^2 - iz = iz + 2 \Leftrightarrow$$

$z^2 - 2iz - 2 = 0; (z-i)^2 - 1^2 = 0; (z-i-1)(z-i+1) = 0$;leads to $z = 1 + I\ or\ z = -1 + i$

Conclusion: The set of points $\boldsymbol{M}$ satisfying $\boldsymbol{M} \equiv \boldsymbol{M}'$ consists of exactly two elements: $\{\boldsymbol{B}, \boldsymbol{C}\}$.

4) Geometric Analysis of Collinearity

a) Proving Collinearity Condition

Three points A, M, and M' are collinear if and only if the vectors $\overrightarrow{AM}$ and $\overrightarrow{AM'}$ are linearly dependent. In the complex plane, this requires the ratio of their corresponding complex vectors to be a purely real number:

$$(z_{M'} - z_A) \,/\, (z_M - z_A) = (z' - i)/(z - i) \in \mathbb{R}$$

From the fundamental relation in part (1), we know that $(z' - i) = 1/(z - i)$. Substituting this into our ratio gives:

$$(z_{M'} - z_A) \,/\, (z_M - z_A) = [\, 1/(z - i)\,]/(z - i) = 1/(z - i)^2$$

For any non-zero fraction $1/W$ to be a real number, its denominator W must also be real. Therefore, $1/(z - i)^2 \in \mathbb{R}$ if and only if $(z - i)^2 \in \mathbb{R}$.

Thus, points $\boldsymbol{A}, \boldsymbol{M}$, and $\boldsymbol{M'}$ are collinear if and only if $(\boldsymbol{z} - \boldsymbol{i})^2$ is a real number.

b) Deducing the Locus of $\boldsymbol{M}$

Let us denote the complex number $(z - i)$ in cartesian form as $X + iY$, where X and Y are real numbers. Geometrically, $X = x$ and $Y = y - 1$, representing the translation of coordinates relative to the point $A(0,1)$.

Expanding the square :

$$(z - i)^2 = (X + iY)^2 = X^2 - Y^2 + 2iXY$$

For this expression to be strictly real, its imaginary component must vanish completely:

$$Im[(z - i)^2] = 0 \Rightarrow 2XY = 0 \Rightarrow X = 0 \;\; or \;\; Y = 0$$

Let us evaluate these two geometric branches:

- Branch 1 ($X = 0$): This means $x = 0$, which corresponds to the y –axis (the imaginary axis).
- Branch 2 ($Y = 0$): This means $y - 1 = 0$, which simplifies to the horizontal line $y = 1$ passing through point A.

Since the transformation is explicitly undefined for $z = i$, point A must be extracted from the final locus set.

Conclusion: The geometric locus of $\boldsymbol{M}$ is the union of the imaginary axis $(\boldsymbol{x} = \boldsymbol{0})$ and the horizontal line $(\boldsymbol{y} = \boldsymbol{1})$, excluding the point $\boldsymbol{A}$.

Second way: A, M, and $M' \Leftrightarrow (z - i)^2 \in \mathbb{R}$

$$\Leftrightarrow 2\arg(z - i) = k\pi, k \in \mathbb{Z} \text{ because } z - i \neq 0$$

$$\Leftrightarrow \left(\overrightarrow{u}; \overrightarrow{AM}\right) = k\pi/2, k \in \mathbb{Z}$$

Conclusion: The geometric locus of $\boldsymbol{M}$ is the union of the two straight lines $(\boldsymbol{OA})$ and $(\boldsymbol{AB})$ excluding the point $\boldsymbol{A}$.

5) Metric and Angular Modulus Interpretations

Starting with the fundamental relationship $(z-i)(z'-i)=1$, we can apply core complex function operations:

• Modulus Rule: Taking the modulus on both sides yields:

$|(z-i)(z'-i)|=|1|$

$$|z-i|\times|z'-i|=1$$

Since $|z-i|$ represents the euclidean distance AM, and $|z'-i|$ represents AM':

$AM\times AM'=1$

• Argument Rule: Taking the principal argument on both sides yields:

$arg[(z-i)(z'-i)]=arg(1)[2\pi]$

$$arg(z-i)+arg(z'-i)=0\ [2\pi]$$

$$arg(z'-i)=-arg(z-i)\ [2\pi]$$

Notice that the vector AB is a horizontal unit vector extending from $A(i)$ to $B(1+i)$, which points precisely along the positive real direction. Therefore, $arg(z-i)$ represents the directed angle $(\overrightarrow{AB};\overrightarrow{AM})$, and $arg(z'-i)$ represents $(\overrightarrow{AB};\overrightarrow{AM'})$. Substituting these gives:

$$\left(\overrightarrow{AB};\overrightarrow{AM'}\right)=\ -\left(\overrightarrow{AB};\overrightarrow{AM}\right)[2\pi].$$

Utilizing directed angle rotation properties, we know that $-(\overrightarrow{AB};\overrightarrow{AM})=(\overrightarrow{AM};\overrightarrow{AB})$. Rearranging the components results in the targeted profile:

$\left(\overrightarrow{AB};\overrightarrow{AM}\right)=\left(\overrightarrow{AM'};\overrightarrow{AB}\right)[2\pi].$

6) Mapping Characterizations for Point $\boldsymbol{K}$

a) **Metric and Angle Measurements for Image $\boldsymbol{K'}$**

We are given that point K has the affix $z_K=i+e^{\mathrm{i}\pi/4}$. Rewriting this gives $z_K-i=e^{\mathrm{i}\pi/4}$.

• Distance $AK=|z_K-i|=|e^{\mathrm{i}\pi/4}|=1$. This proves K is situated exactly on circle (C).

• Angle $\left(\overrightarrow{AB};\overrightarrow{AK}\right)=arg(z_K-i)=\pi/4$ (or 45°).

Applying the specific mapping transformations deduced in part (5) directly to point K:

• Distance calculation: $AK\times AK'=1\Rightarrow 1\times AK'=1\Rightarrow AK'=1$.

• Angular calculation: $(\overrightarrow{AB};\ \overrightarrow{AK'})=-\left(\overrightarrow{AB};\overrightarrow{AK}\right)=-\pi/4\ (or-45°)$.

Summary: The distance $\boldsymbol{AK'=1}$, and the directed angle measure $\boldsymbol{(\overrightarrow{AB};\ \overrightarrow{AK'})=-\pi/4}$.

b) Graphical Plotting Guide for $\boldsymbol{K'}$

We have $AK=AK'$ and $(\overrightarrow{AB};\ \overrightarrow{AK})=-\left(\overrightarrow{AB};\overrightarrow{AK'}\right)$ then, K' is the orthogonal symmetry of K with respect to the straight line (AB).

c) Locus Transformation of the Semi-Straight Line

When M traces the semi-straight line $[AK)$ excluding point A, the angle is permanently fixed at

$(\overrightarrow{AB};\overrightarrow{AM}) = \pi/4$.

Consequently, its image point M' must maintain a fixed angular path satisfying $(\overrightarrow{AM'};\overrightarrow{AB}) = -\pi/4$.

As M moves outward from point A towards infinity (meaning the length AM spans from 0 to ∞), the inverse relationship dictating the distance requires $AM' = 1/AM$ to span from ∞ down towards 0 (approaching but never reaching A).

Conclusion: The set of image points M' traces the semi-straight line $[AK')$ pointing at an angle of $-45°$, excluding the boundary point A.

7) Geometric Projection and Inversion Properties

a) Metric Inversion Proof: $AQ \times AE = 1$.

Based on the provided geometric configuration, points K and Q share an argument of $\pi/4$ relative to A. Thus, A, K, E, and Q are strictly collinear along the same ray line.

We are given that triangle AHQ is a right-angled triangle at H ($\angle AHQ = 90°$), and E is the orthogonal projection of vertex H onto the hypotenuse ray line AQ. This setup allows us to apply the right-triangle Euclidean geometric projection theorem:

$$AH^2 = AE \times AQ$$

Because H is defined as an element on the circle (C) which has a fixed radius of 1 centered at A, the segment length AH equals 1. Substituting this radius value gives:

$$1^2 = AE \times AQ \Rightarrow AQ \times AE = 1.$$

Hence, the metric inversion property is proven: $AQ \times AE = 1$.

b) Precise Geometrical Construction of Q'

First way:

We have, $AE \times AQ = 1$ and $AQ \times AQ' = 1$, then $AE = AQ'$ this leads to: Q' belongs to the circle (S) of center A and radius AE.

On the other hand, Q' belongs to the semi line $[AK')$ except A

We can conclude that $\{Q'\} = (S) \cap [AK') - \{A\}$